CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

Topoisomerase: 9 molecules stopped early

13 stopped trials across 9 molecules: Irinotecan, Doxorubicin, Etoposide, Idarubicin and 5 more. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

3.9%
13 of 335 closed trials (95% CI 2.3%–6.5%)

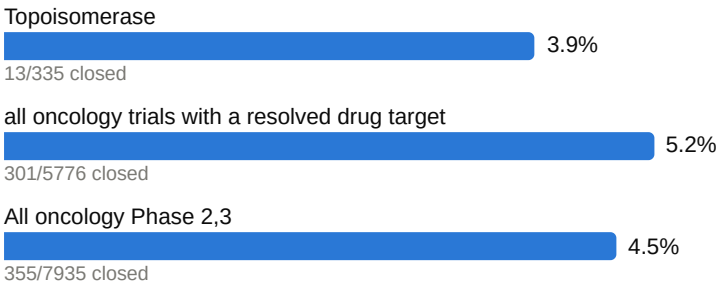
9 / 2 / 1
efficacy / safety / both (adds to 13)

29.7%
of 1128 trials have closed · 793 still open or unresolved

0.6%
stopped for a biological reason within 3 years of starting (95% CI 0.3%–1.5%; 537 still at risk; 1.5% area-wide)

5 of 8 **Completed and missed the primary endpoint.** Of 236 completed trials in this segment, 8 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 5 missed and 3 met. Counted separately and never in the rate above; a floor, not a rate. 92 of the 218 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 517 distinct experimental drugs across 1128 trials (small molecule 950, antibody 468, adc 287, unknown 200). Most tested: IRINOTECAN (313), DOXORUBICIN (275), ETOPOSIDE (224), CYCLOPHOSPHAMIDE ANHYDROUS (206), FLUOROURACIL (165). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 13 stops came from 13 sponsor-asset programmes across 13 sponsors; the largest (University of Nebraska) contributed 1; removing that programme's trials from both sides leaves 3.6%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 21 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 10.1%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	5	156	3.2%	1.4%–7.3%
2018–2020	7	106	6.6%	3.2%–13.0%
2021–2024	1	73	1.4%	0.2%–7.4%

Where the stops sit

Efficacy / futility: 9, Safety: 2, Efficacy and safety: 1, Benefit–risk: 1. Sponsors with most stops: University of Nebraska (1), Roche (1), Memorial Sloan Kettering Cancer Center (1), Sanofi (1), UNICANCER (1), Daiichi Sankyo (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04141995	2	University of Nebraska	DIGOXIN, FLUOROURACIL, CALCIUM, IRINOTECAN	2025-05	Efficacy	Due to low accrual and futility
NCT04029688	2	Roche	IDASANUTLIN, VENETOCLAX, CYCLOPHOSPHAMIDE ANHYDROUS, TOPOTECAN	2024-05	Efficacy + safety	Study was terminated by the sponsor based on insufficient tolerability and efficacy to proceed beyond Part 1b.
NCT03879798	2	Memorial Sloan Kettering Cancer Center	VALEMETOSTAT, IRINOTECAN	2023-05	Safety	Due to DLTs
NCT03860844	2	Sanofi SNY	ISATUXIMAB, DEXAMETHASONE, FLUDARABINE, CYTARABINE	2022-09	Efficacy	Study was prematurely stopped due to sponsor decision (stage 2 efficacy criteria not met); not due to safety concerns.
NCT03722108	2	UNICANCER	REGORAFENIB, IRINOTECAN	2022-05	Benefit–risk	Unfavorable benefit-risk balance in the experimental arm following the IDMC for intermediate efficacy and safety analysis
NCT04479436	2	Daiichi Sankyo	PATRITUMAB DERUXTECAN	2022-02	Efficacy	Study was terminated early given the Interim Analysis for Part 1 (signal finding) did not meet pre-specified criteria and will not proceed to Part 2. Sponsor will proceed closing the study.

The 6 most recent of 13 stopped trials. Every molecule behind all 13 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

Method
Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 29.7% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.